Coding Sample: Longitudinal Mixed-Effects Modeling in R

Repository: https://github.com/syp4137/regression-analysis-coding-sample.git

Project Overview

This coding sample demonstrates a longitudinal modeling workflow in R using linear mixed-effects models. The analysis examines how body-fat trajectories change before and after menarche, using repeated measurements from adolescent girls. I selected this sample because it reflects how I approach hierarchical data: structuring time carefully, fitting interpretable mixed models, and comparing random-effects specifications to determine which sources of variation are supported by the data.

- **Language:** R
- **Dataset:** MIT Growth and Development Study (Phillips data)
- **Focus:** longitudinal data, piecewise time effects, mixed-effects modeling, random-effects comparison, interpretation

Data and Objective

The dataset contains repeated body-fat measurements from 162 girls observed over time. For each subject, the timing of menarche is known, allowing age to be centered relative to that event. Following Laird and Fitzmaurice, I model the trajectory of body fat separately before and after menarche using two time variables:

- `t_minus`: adjusted age before menarche, and 0 afterward
- `t_plus`: adjusted age after menarche, and 0 before

Objective: Model body-fat trajectories before and after menarche, and determine which subject-specific random effects are needed to capture individual variation over time.

Data Preparation

I first prepared the piecewise time variables used in the mixed model.

```
library(lme4)
library(dplyr)

phillps <- read.table("Phillips.txt", header = TRUE)

phillps <- phillps %>%
  mutate(
    t_minus = ifelse(age.adjusted < 0, age.adjusted, 0),
    t_plus  = ifelse(age.adjusted >= 0, age.adjusted, 0)
  )

str(phillps)
```

```
'data.frame':   1049 obs. of  7 variables:
 $ subject     : int  1 1 1 1 1 1 2 2 2 2 ...
 $ age         : num  9.32 10.33 11.24 12.19 13.24 ...
 $ menarche    : num  13.2 13.2 13.2 13.2 13.2 ...
 $ age.adjusted: num  -3.87 -2.86 -1.95 -1 0.05 1.05 -4.44 -3.2 -2.25 -0.51 ...
 $ body.fat    : num  7.94 15.65 13.51 23.23 10.52 ...
 $ t_minus     : num  -3.87 -2.86 -1.95 -1 0 0 -4.44 -3.2 -2.25 -0.51 ...
 $ t_plus      : num  0 0 0 0 0.05 1.05 0 0 0 0 ...
```

```
summary(phillps[, c("body.fat", "age.adjusted", "t_minus", "t_plus")])
```

```
    body.fat      age.adjusted        t_minus            t_plus
 Min.   : 2.11   Min.   :-6.3100   Min.   :-6.3100   Min.   :0.000
 1st Qu.:18.24   1st Qu.:-1.6100   1st Qu.:-1.6100   1st Qu.:0.000
 Median :23.59   Median : 0.1500   Median : 0.0000   Median :0.150
 Mean   :23.61   Mean   : 0.1892   Mean   :-0.8866   Mean   :1.076
 3rd Qu.:29.56   3rd Qu.: 2.0500   3rd Qu.: 0.0000   3rd Qu.:2.050
 Max.   :45.49   Max.   : 4.6300   Max.   : 0.0000   Max.   :4.630
```

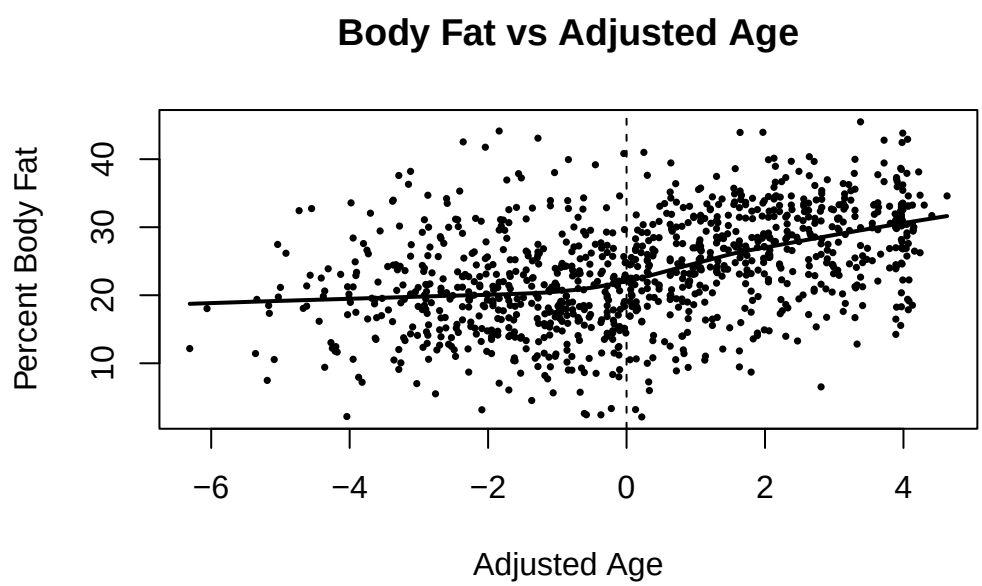
This parameterization allows the model to estimate different average slopes before and after menarche while preserving a common intercept at the event time.

Exploratory Analysis

To examine the overall longitudinal pattern, I plotted body fat against adjusted age and added a LOWESS smooth.

```r
plot(phillps$age.adjusted, phillps$body.fat,
     xlab = "Adjusted Age",
     ylab = "Percent Body Fat",
     main = "Body Fat vs Adjusted Age",
     pch = 16, cex = 0.5)

lines(lowess(phillps$age.adjusted, phillps$body.fat), lwd = 2)
abline(v = 0, lty = 2)
```



The overall trend suggested a changing trajectory around menarche, supporting a piecewise longitudinal model rather than a single linear time effect.

Mixed-Effects Models

I first fit the full mixed-effects model with random intercepts and separate random slopes for the pre- and post-menarche time variables.

```r
m_full <- lmer(
  body.fat ~ t_minus + t_plus + (1 + t_minus + t_plus | subject),
  data = phillps,
  REML = FALSE
)
```

I then fit reduced models to test whether both subject-specific slopes were necessary.

```r
m_no_plus <- lmer(
  body.fat ~ t_minus + t_plus + (1 + t_minus | subject),
  data = phillps,
  REML = FALSE
)

m_no_minus <- lmer(
  body.fat ~ t_minus + t_plus + (1 + t_plus | subject),
  data = phillps,
  REML = FALSE
)

m_int <- lmer(
  body.fat ~ t_minus + t_plus + (1 | subject),
  data = phillps,
  REML = FALSE
)
```

These models allow a structured comparison of whether individual girls vary primarily in baseline body fat, in their pre-menarche slope, in their post-menarche slope, or in all three.

Model Comparison

I compared the nested mixed models using likelihood-ratio tests.

```r
anova(m_full, m_no_plus)
```

```
Data: phillps
Models:
m_no_plus: body.fat ~ t_minus + t_plus + (1 + t_minus | subject)
m_full: body.fat ~ t_minus + t_plus + (1 + t_minus + t_plus | subject)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
m_no_plus    7 6110.7 6145.4 -3048.3    6096.7
m_full      10 6078.3 6127.9 -3029.2    6058.3 38.393  3  2.333e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(m_full, m_no_minus)
```

```
Data: phillps
Models:
m_no_minus: body.fat ~ t_minus + t_plus + (1 + t_plus | subject)
m_full: body.fat ~ t_minus + t_plus + (1 + t_minus + t_plus | subject)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
m_no_minus    7 6119.6 6154.3 -3052.8    6105.6
```

```
m_full       10 6078.3 6127.9 -3029.2    6058.3 47.278  3  3.033e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(m_no_plus, m_int)
```

```
Data: phillps
Models:
m_int: body.fat ~ t_minus + t_plus + (1 | subject)
m_no_plus: body.fat ~ t_minus + t_plus + (1 + t_minus | subject)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
m_int        5 6160.2 6185.0 -3075.1    6150.2
m_no_plus    7 6110.7 6145.4 -3048.3    6096.7 53.505  2  2.407e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(m_no_minus, m_int)
```

```
Data: phillps
Models:
m_int: body.fat ~ t_minus + t_plus + (1 | subject)
m_no_minus: body.fat ~ t_minus + t_plus + (1 + t_plus | subject)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
m_int         5 6160.2 6185.0 -3075.1    6150.2
m_no_minus    7 6119.6 6154.3 -3052.8    6105.6 44.621  2  2.045e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The full model fit substantially better than models that removed either the random post-menarche slope or the random pre-menarche slope. In particular, removing the post-menarche random slope significantly worsened fit, and removing the pre-menarche random slope also significantly worsened fit. These comparisons indicate that girls differ not only in baseline body-fat levels, but also in how their trajectories evolve both before and after menarche.

Interpreting the Preferred Model

After selecting the full model, I examined the fixed effects and variance components.

```r
summary(m_full)
```

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: body.fat ~ t_minus + t_plus + (1 + t_minus + t_plus | subject)
   Data: phillps

      AIC       BIC    logLik -2*log(L)  df.resid
   6078.3    6127.9   -3029.2    6058.3      1039
```

```
Scaled residuals:
    Min      1Q  Median      3Q     Max 
-2.7817 -0.5940 -0.0368  0.5933  3.3821 

Random effects:
 Groups   Name        Variance Std.Dev. Corr       
 subject  (Intercept) 45.6180  6.7541              
          t_minus      1.6018  1.2656    0.29      
          t_plus       0.8635  0.9292   -0.56 -0.09
 Residual              9.4768  3.0784              
Number of obs: 1049, groups:  subject, 162

Fixed effects:
            Estimate Std. Error t value
(Intercept)  21.3614     0.5628  37.958
t_minus       0.4171     0.1564   2.667
t_plus        2.4640     0.1187  20.750

Correlation of Fixed Effects:
        (Intr) t_mins
t_minus  0.351       
t_plus  -0.521 -0.348
```

```r
VarCorr(m_full)
```

```
 Groups   Name        Std.Dev. Corr       
 subject  (Intercept) 6.75411             
          t_minus     1.26564   0.291     
          t_plus      0.92923  -0.565 -0.095
 Residual             3.07843             
```

The fixed-effects structure estimates separate average slopes before and after menarche, while the random-effects structure allows each subject to deviate from the average intercept and both average slopes. This makes the model both substantively meaningful and statistically appropriate for repeated measurements collected at irregular ages.

Diagnostic Checks

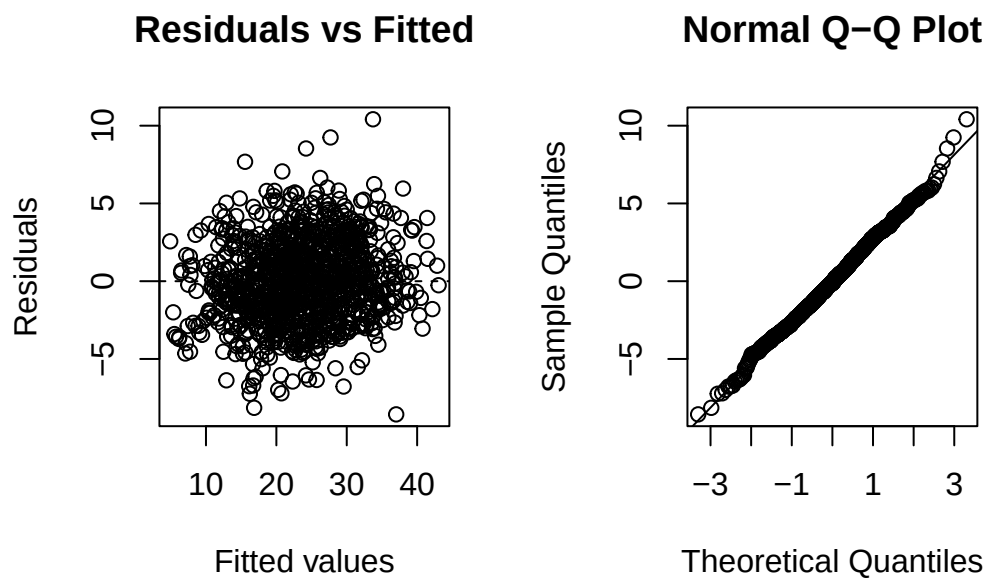
Finally, I examined residual diagnostics for the preferred mixed model.

```r
par(mfrow = c(1, 2))

plot(fitted(m_full), resid(m_full),
     xlab = "Fitted values",
     ylab = "Residuals",
```

```
     main = "Residuals vs Fitted")
abline(h = 0, lty = 2)

qqnorm(resid(m_full))
qqline(resid(m_full))
```




```
par(mfrow = c(1, 1))
```

The diagnostics help assess whether the selected model captures the main longitudinal structure without leaving strong residual patterns unexplained.

Key Takeaways

This sample highlights the kind of modeling work I most enjoy: structuring repeated-measures data carefully, fitting interpretable mixed-effects models, and comparing competing random-effects structures rather than assuming a default specification. In this analysis, the strongest evidence supported a model with subject-specific baseline levels and separate subject-specific slopes before and after menarche, reflecting meaningful individual heterogeneity in developmental trajectories.